Frankie: current models and selection systems

Dated 11 October 2026. Prepared for Dave.

This handbook describes what currently produces the followed book, what produces the separately delivered shadow, and what is research only. It is based on today's dated artifacts, frozen model manifests and inspected execution code—not on the names of old files or a proposed future design. It does not change any model or rule.

1. The essential distinction

We do not have one model with three universally applied steps. We have several separate lanes.

Lane	Purpose	Appears in normal followed total?
Core Production	Existing model candidates, confidence, pace and comparability rules	Yes
Followed add-ons	Benter No-Current, Bred Signal, Top Combo, Weighted, PR1 agreement	Yes, after merging and filtering
Delivered Flat Step-1 shadow	Monthly LightGBM full-field rankings, then source agreement gates	No
Delivered Jumps Step-1 shadow	Monthly CatBoost full-field rankings, then source agreement gates	No
Controls and prospective challengers	Measure alternative recipes against references	No
Own rating, Step-3 and residual suitability experiments	Test new past-performance ideas	No

Step-1 usually refers to the separate rolling-rich shadow model, not the entire followed book. A followed selection can qualify through Benter or a PR 1 agreement without being a Step-1 shadow selection. Conversely a shadow selection need not qualify for followed Production.

Calling a result a model score, a probability, a value flag and a bet are four different things. A row in a scored universe is not a final bet. The final followed artifact is combined_followed_with_blend_YYYY-MM-DD.csv; the delivered shadow artifact is rolling_rich_step1_shadow_YYYY-MM-DD.csv.

2. Data sources and their roles

Daily workbook / RBD

The Daily workbook and archived model sheets are authoritative for the established workflow. They supply runner identity, declared race conditions, horse attributes, OR, declared weight, source pace, prior-run summary fields and the established model context. They are not replaced by a new freeform selection opinion. RBD native ratings and ranks are also available for research, but are not extra inputs to the currently delivered Step-1 model.

Rating The Races (RTR)

RTR supplies prediction-time Win Odds, its score/rank, signal counts and its statistical fields. From 4 October the followed book uses validated **RTR Win Odds as its decision price**: floors/ceilings, edge recomputation,

Weighted tie-breaking, PR 1 agreement price eligibility and final displayed price. This price-source switch preserves the established model score and source-native signal scores.

Later Industry SP and BSP are settlement inputs, not today's decision quote. A legacy column named Industry SP in a followed candidate file can hold the decision-time RTR quote; its name alone is not proof of which price was used. decision_price_source and rtr_decision_odds are the audit columns. Missing, ambiguous or result-enriched decision quotes are not silently replaced in the final followed book.

Racedata / RaceAdvisor

Prediction-time PR Score, PR Rank, Dynamic speed rating and Race Predictor fields support different lanes. PR Score and PR Rank enter current Step-1; raw Dynamic speed enters Flat Step-1 only. Race Predictor Win% enters the core confidence check, but is not an additional current Step-1 feature and was removed from the followed Benter probability blend. PR Rank also acts as a later selection gate in specified lanes.

RaceIQ

Measured sectionals, finishing speed and stride evidence are used for lagged contextual analysis and post-race pace checks. Prior runs can supply evidence before today's race. Today's actual sectionals are post-race truth and cannot be used in today's forecast. **RaceIQ measured sectionals are not inputs to the currently delivered Step-1 models.** Source Pace in Step-1 is not the same thing as a measured RaceIQ sectional profile.

Historical results archive

The multi-year archive supports prior-performance ledgers, draw/pace context and the experimental own rating. Five years of results does not mean five years of complete contemporaneous Step-1 feature snapshots. A historical results export cannot supply missing morning quotes or prediction-time ranks retroactively.

Inputs are frozen separately under data/source_sets/pre_race/YYYY-MM-DD/ and post_race/YYYY-MM-DD/. Raw sources and issued artifacts remain preserved. Hashes, source-set IDs and pre-off seals provide the audit trail.

3. Core followed Production

3.1 Existing scoring workflows

The established Flat handicap workflow uses a LightGBM binary winner model with 45 price-aware inputs. Its no-price companion removes six explicit market fields, leaving 39 inputs. The shared 45 include horse/race attributes, OR and weight relative to the field, source pace, market context, course/going/race type, connections/equipment identities and course/distance-winner flags.

These are legacy production workflows, distinct from the new monthly47/48-input Step-1 shadow. Legacy fits use the supplied historical training source; their training policy must not be described as the rolling Step-1 strictly previous-month policy. Scores are model estimates, not a guarantee of well-calibrated or 100%-summing race probabilities.

The Jumps legacy base is also a LightGBM binary workflow. Its established suitability component uses 25 summary inputs, including latest finishing position, last-five wins, activity, course/distance/going wins and derived summary ratios. The blend is **80%base prediction +20%suitability prediction**. The suitability component uses five race-grouped fitting folds and isotonic mapping; that grouped calibration is not the chronological OOF procedure used in newer research challengers.

The 25-input Jumps suitability list is in Appendix B. **Some past-performance summaries are already used in legacy Jumps Production; that does not mean they are inputs to the separately delivered Jumps CatBoost shadow.**

3.2 Candidate universe

Core candidates are the de-duplicated union of the capped-control and legacy uncapped/core books. The capped control is not itself the followed book.

Flat candidate generation retains the established model/price/edge and no-price-agreement or premium-quality process. Relevant entry rules include:

- Capped path: model rank at most 2, decision odds at least 3 and below 8, initial probability-edge floor 0.05; retained higher-quality candidates then require edge at least 0.10.

- Uncapped path: decision odds at least 3, initial model rank at most 2, edge at least 0.20; after the existing candidate tie-break, retain rank 1 only, without backfilling a different runner when the tie-break selected rank 2.

- Legacy Flat quality screening keeps candidates meeting the established agreement/premium requirements; this is not a free selection from every scored runner.

Core edge = model probability −1/decision odds. This is a difference in probability, not Benter's probability-times-price return ratio. Confusing 0.20 core edge with 20%relative value materially changes a rule.

Some recency checks remain in earlier legacy candidate generation (for example the uncapped Flat 180-day screen and legacy Jumps 90-day path). Removing the core 60-day confidence/rescue veto does **not** imply that every recency check in every upstream workflow disappeared. See source code and dated candidate audit for a runner's actual path.

3.3 Current five-check confidence gate

Effective 4 October, score one point for each:

1. Prediction-time Racedata PR Rank 1–3.

2. Prediction-time RTR Rank 1–3.

3. Daily model/Production race rank 1–3.

4. Dynamic speed rating at least 190.

5. Race Predictor Win%at least 15.

Normal pass is **at least 3 of 5**. This is an eligibility score, not a 30%,60%or 100%win chance. Days Since Last Ran no longer scores confidence; the old core/rescue60-day veto and old6/6race-relative add-back are retired for current cards. Historical issued cards retain their historical rules.

Exceptions still exist: qualifying Jumps pace-neutral/pace-okay candidates can use the retained add-back. Otherwise-dropped core candidates can use the Bred rescue at 1 pt: inspected code requires Bred rank 1, score at least 5 and gap at least 3. Rescue does not manufacture a new horse outside the core candidate universe. Top Combo and Weighted are not substitutes for this rescue.

3.4 Pace and comparability

- Core pace advantage:3 pt; pace pressure risk:1 pt; level/neutral/other:2pt. Bred rescue overrides to 1 pt.

- Flat uncapped-only candidates at decision odds 20 or more are vetoed when pace is pressure-risk.

- Flat core must be **Proven comparable** after its other gates.

- Comparability uses strictly earlier top-three runs, same code/surface, approximately comparable distance, at least six usable dimensions, at least 67%agreement and evidence within two years.

Comparability means there is qualifying evidence under the rule, not a calibrated probability. Failure can mean unsuitable or insufficient evidence; it is not proof that the horse cannot handle today's conditions.

4. Followed add-ons

These are rule-based selection lanes, not necessarily independently trained machine-learning models.

4.1 Benter No-Current — followed

GB Flat/all-weather handicaps. This is our named score/value engine, **not a claim to reproduce Bill Benter's proprietary model**.

Current blend:

- Two-thirds RTR score-derived race probability (softmax).

- One-third RTR signal-count-derived race probability (normalized within race).

- Normalize the resulting field; fair odds=1/probability.

Daily Race Predictor probability is absent from this blend from 26 September. Normalized probabilities summing 100%do not by themselves establish calibration.

Base selection gates: decision odds 3–8 inclusive, model rank at most 2, agreement count at least 2, relative value edge at least 0.10, and one selected candidate per race under the engine's policy. Benter relative edge=probability×odds−1. Its agreement count combines modeltop 3, RTRtop 3, RTRsignal-count score at least 3 and RTRValue%at least 110.

The no-current-race rule excludes races already occupied by the core/current-followed input at that stage. **It is not a promise that the final merged book has only one horse per race**: later add-ons can occupy the same race.

Racedata PR must be 1–3 from 27 September. Missing PR match/rank requires review, not an invisible deletion. Comparability is recorded but does not filter this followed lane. Base stake 2 pt for agreement count 4, otherwise 1 pt; eligible final Flat angle uplift can change stake.

The full No-Current qualifier set is followed. Do not apply the Hybrid tightening to it just because an artifact name contains “action”.

4.2 Benter Hybrid — comparison only

The action shortlist adds model rank 1, pre-filter Production rank at most 3 and Benter relative edge at least 0.25. It remains a comparison view, not an extra followed book. combined_followed_with_benter_action_YYYY-MM-DD.csv is therefore **not** the ordinary followed artifact.

4.3 Bred Signal — followed

An RTR sire-statistics score using evidence for distance/surface, trip, course-distance and going. It is not our new own achieved-performance rating.

Standalone gates: handicap, Bred race rank 1, gap to second at least 3, Bred score at least 5, RTRValue%at least 125 and decision odds at least 4. No promoted upper odds cap applies. The≤20 monitor is research only.

Base stakes:1 pt;2 pt at Value%≥150;3 pt at Value%≥225. The score awards positive/negative points to sufficiently sampled sire A/E and win/place statistics; thresholds and points are documented in source tools/backtest_rtr_bred_for_trip_top5.py. A/E compares actual wins with market-expected wins; it is not a horse's individual win probability.

4.4 Top Combo — followed

An RTR trainer/jockey statistical score: direct combination record is strongest, with recent and course/trip/distance/surface support.

Gates: handicap, combo rank 1, decision odds 4–10 inclusive and Racedata PR 1–3. Standard tier: Value%≥100 and score gap≥4. Near-miss: Value%85 to below 100 and gap≥5, at 1 pt.

Standard stake 1 pt;2 pt when odds≥10(which now means exactly 10 under the cap);3 pt when Value%≥200. Near-miss stays 1 pt before any permitted book-level handling. This is not the same thing as Jockey and Trainer identity categories in Step-1.

4.5 Weighted To Win v 2 — followed

Handicap, at least 4 runners, decision odds 3–10 inclusive, score≥3, race rank 1, at least 2 prior OR runs, a prior win at today's OR or higher, latest numeric form figure≤6;0 means 10 th-or-worse. If several qualifiers tie, keep the shortest validated RTR-priced qualifier in that race.

Score: prior win at/higher today's OR+2; prior winning OR at least 3 higher+1; prior top-three at/higher today's OR+1; prior top-three OR at least 3 higher+1; last OR at least 3 higher+1; last declared weight at least 3 lb higher+1; fewer than 2 prior OR runs−1.

Base stake 1 pt. Proven-comparable Flat qualifier gets+1 pt; Jumps qualifier is followed only if Proven comparable. This rule uses official marks as historical handicapping evidence; it does not claim a literal equal-weight counterfactual.

4.6 PR 1 agreement — followed

GB Flat handicap add-on from the final filtered Racedata PR 1 source candidates, with validated decision odds≥3 and either RTRrank 1 or pre-filter Production rank 1. Stake 3 pt. If both agreements hold, it is one runner, not two bets. Its overlap with another signal is also merged, not added to create 6 pt.

Racedata-native PR 1 is a separate control; following this agreement does not promote the entire PR 1 control book.

4.7 Final merging and stake context

Same horse across lanes: retain once, combine signal labels and take the highest applicable stake rather than adding stakes.

Flat runners with at least two live angle hits can get+1 pt only in strict Quality Class Drop context: class drop, latest numeric form≤6 and the applicable price check≤20; maximum 3 pt. Jumps has no angle-combination uplift. Hot Race, Quality Class Drop and Calibrated Score do not create standalone bets. The 15 ptdaily target is only a shadow comparison, not compulsory staking.

Final followed race guard: at most 2 distinct selections. If a race has more than 2 after merging, the code removes all its followed rows; it does not simply choose the best 2. This differs from both the Flat-shadow exactly-one rule and the Jumps-shadow keep-both rule.

Flat Blend Extra remains an empty/compatibility lane in the inspected current builder; an old function and filename do not mean it is an active additional selector.

5. Delivered rolling-rich Step-1 shadow

5.1 The two models

Item	Flat	Jumps
Architecture	LightGBM binary winner model	CatBoost YetiRank race-level model
Current inputs	48	47
Current training policy	Strict previous calendar month	Strict previous calendar month
October fitted training cohort	115 races / 1,203 runners, 2–24 September	71 races / 484 runners, 4–29 September
Additional raw Dynamic speed	Yes, from6October	No
Output delivered to selection	Full-field uncalibrated score and rank	Full-field uncalibrated score and rank

Both score the complete eligible handicap field before gates. LightGBM learns nonlinear splits/interactions to predict a binary outcome; CatBoost's ranking objective learns ordering within races. The inspected Flat recipe uses 300 boosting rounds, learning rate 0.05 and 31 leaves. Jumps uses 300 iterations, depth 4, learning rate 0.05, L 2 leaf regularization 5 and seed 42. These are training settings, not betting stakes.

Neither has fixed human-assigned weights such as “OR=20 points”. Feature importance is not a complete independent contribution or a causal effect.

Current monthly models are frozen with features and hashes. Missing/incomplete source data or the required month model must not silently substitute fixtures or stale evidence. The separate original trial runs 5 October 2026–4 April 2027, then review. Model changes and issued-card rules are versioned separately.

5.2 Current inputs by function

The complete 48-input dictionary is Appendix A. Jumps has the same list except dynamic_speed.

- Race/conditions:8inputs.

- Age/experience/recency:9.

- OR/field-relative OR:7.

- Declared weight/field-relative weight:4.

- Source pace/field-relative pace:5.

- RTR morning-market context:6.

- Connections/equipment/course-distance flags:6.

- Racedata rating features:3 Flat,2 Jumps.

industry_prob is a legacy name: in the daily Step-1 build it is1/validated RTR morning Win Odds. Top-two/top-three market “share” fields are raw sums of implied probabilities, not normalized percentage shares. pr_score_log=log 1 p(nonnegative PR Score). Dynamic speed is raw prediction-time Racedata speed, not RaceIQ speed.

5.3 Step-2 selection gates

1. Rank all runners within the full field by Step-1 score.

2. Consider model top 2 only.

3. Each must have prediction-time RTRrank 1–3 and Racedata PRrank 1–3.

4. **Flat:** count qualifying horses before price. If exactly 1 passes, retain it; if 2 pass, exclude both.

5. **Jumps:** retain both when both pass.

6. Apply validated RTR odds≥3 to each retained candidate, then 1 ptwin. No upper cap or additional value gate.

The≥3 floor was explicitly amended pre-off on 6 October and is present in today's provenance. Older monthly manifest text still says “both retained/no price gate”; that text describes the original fit-era rule, not today's full selection policy. **Use today's provenance and amendment for current gates.** Counting before price prevents a short-priced runner's deletion from turning a two-qualifier Flat race into a one-qualifier race.

5.4 What the shadow does not currently do

No automatic Step-3 achieved-performance points, own opponent-network rating, residual conditions suitability, seasonal horse preference, historical draw correction, RaceIQ measured sectionals, explicit trainer/jockey strike rates, extra RTRScore or RBDRating input, or Daily Race Predictor probability input. Some are research challengers; none becomes live merely because its test returned profit.

RTRrank is an external gate, not a current Step-1 predictor. PRrank is both a predictor and external gate. OR and declared weight are separate inputs; OR is an ability/handicap mark, not identical to weight. Out-of-handicap runners, claims, allowances and race conditions can complicate that relationship.

6. Race conditions and past performance: what is actually represented

Track and Going are categories; Class, distance and field size provide numeric/context inputs. The model can learn interactions between these and horse attributes, but a condition shared by all runners cannot by itself identify which particular horse prefers it.

Course Winner and Distance Winner indicate historical summary success. They do not measure how far above its own baseline the horse performed, consistency, evidence strength or opponent quality. Jockey/Trainer identities can learn patterns but are not an explicit recent strike-rate calculation. Source Pace and front-runner count describe anticipated shape, not a calibrated forecast of measured sections.

Legacy Jumps Production includes more explicit past-form/win-count summaries than the delivered Step-1 shadow. Core comparability and Weighted also consult prior runs. Therefore “we use no past performance” is wrong, but **we do not yet have a proven, detailed horse-level achieved-performance suitability block in the delivered Step-1 models**.

7. Research models and controls we must keep separate

7.1 Established controls

- Capped-control, legacy uncapped and no-price/Flat controls: preserve comparators and old behavior; not separate followed totals.

- Racedata-native PR 1 Flat control: its native source odds and own value-based stake comparison; not the currently displayed daily shadow.

- Previous Flat CatBoost and new-LightGBM Step-1 controls: retained for historical comparison; not replacements for today's rolling-rich delivered model.

- Benter Hybrid action view: comparison only.

- Calibration, angle-guard and Racedata-top-three/rank-sum forward artifacts: monitoring/comparison outputs, not standalone permission to bet.

“followed” in an old compatibility filename is not sufficient to promote a control. Scope, provenance and the final standard followed artifact determine its role.

7.2 Six-month Step-1 tournament

Registered 11 October 2026–10 April 2027, reviews 11 January and 11 April. Initial scope is GB Flat handicap full fields, not new Jumps challengers.

Five fixed recipes: reconstructed unchanged Step-1 reference, extra Race Predictor fields, extra RBD fields, both feature groups, and market-only reference. The two group labels mean **base features plus that group**, not models using only four columns. Market-only is the exception.

Models/calibration refresh from complete resolved sealed previous-month captures; October starts from fixed September-fit research versions. Full-field loss/Brier/calibration are primary; all probability×frozen RTRodds>1 qualifiers at 1 ptare secondary. No source-rank gates or delivered-shadow single-qualifier rules should be assumed for those value books. Arms overlap, so their P/L cannot be added.

Operational status inspected today at 08:43 UTC:0 captured dates,0 settled,1 blocked date; reason RESULT_ENRICHED_SOURCE. Being registered/wired is not the same as collecting a valid prospective observation. This handbook does not claim the block was repaired.

7.3 Separate fixed rating-feature trial

Registered 11 October–8 January, review 9 January. Similar five arms, but holds its September-fit October research models fixed instead of refreshing monthly. It is a different registration and must not be merged with the six-month tournament. Its registration says armed/zero prospective samples; no verified successful capture is claimed here.

7.4 Draw/pace context trial

10 October–7 January, review 8 January. Annotation/paper-veto comparison, separate RaceIQ checking; not a model replacement. At inspection:2 captured days,19 full-field races,11 baseline nominations,1 settled day, no vetoes;9 active baseline and paper bets, both−9 ptgross BSP;2 measured RaceIQ observations. That is too little prospective evidence for an efficacy claim.

8. Our own past-performance rating

Separate OR/price-independent opponent-network prototype. The multi-year archive warms the opponent-strength network; monthly ratings and each achieved run use strictly earlier evidence. Stable 100 reference is an arbitrary rating-scale convention, **not an official 100 OR or 100%chance**.

An achieved run compares the horse's declared-weight/beaten-length-adjusted performance with sufficiently supported earlier-rated rivals. The present conversion uses fixed experimental weight/length assumptions (including distance-dependent length conversion), not a validated causal estimate of what would happen if all horses carried equal weight. Claims/net weight and timing/pace corrections are not fully established.

Recent ability, comparable-condition mean, variability, evidence count and recency form prediction-time profiles. Supported coverage increased after the full archive integration; it is still not universal. Cold starts, too few usable performances and disconnected/unsupported opponent evidence remain explicit.

Step-1 own-rating-feature and standalone rating tests are completed research, not current live inputs. Source ledger: reports/frankie_own_performance_rating/archive_v2_2026-10-10/. The stand-alone and Step-3 chooser reports document their specific gates and restricted cohorts, which differ from full-field Step-1.

9. New residual-suitability challenger: latest completed result

This measures achieved historical figure minus what the horse was expected to achieve **before that run**. Smooth going/log-distance kernels and recency weight earlier residuals; a condition-specific contrast is shrunk toward neutral. Evidence mass, Kish effective count, variability/uncalibrated uncertainty proxy, similar-condition peak and recency are recorded. It is not the older achieved-minus-OR experiment.

The challenger has 48 current research baseline inputs +8 absolute-own-rating features +21 suitability features. Parameters tuned on 9,168 chronological 2025 development runs; no 2026 tuning. Complete matched probability comparison:527 races /4,919starters, August–9 October.

- Unchanged reference winner loss 2.077984; absolute own-rating control 2.076830; suitability 2.081585. Lower is better: no demonstrated overall accuracy improvement.

- Fixed research top2/RTR/PR/exactly-one paper gate, **without the later live price-floor amendment**: baseline277ran/66wins/+0.09ptgross BSP; suitability283ran/74wins/+25.42pt. The paired gain interval includes no gain, so this is not a validated betting edge.

- Larger fixed going/trip component diagnostic:20,226 historical achieved runs across 3,756 represented races, January–8 October. Adding the joint conditions correction slightly worsened performance-forecast error. This is **not** a 3,756 complete-race Step-1 probability or P/L backtest.

Underlying figures are experimental, archives mostly retrospective, and 2026 research is heavily reused. No promotion or live change follows from either result. Reports: reports/horse_residual_suitability/2026-10-11/REPORT.md and larger_2026_v1/REPORT.md.

10. How performance is measured

Forecast quality

Winner log-loss penalizes giving the winner too little probability; runner Brier measures squared probability error; calibration compares predicted frequencies with realized frequencies; top-one/top-two hit rates assess ordering. Research probability models normalize whole fields; normalization alone does not ensure useful calibration. Current delivered raw Step-1 scores must not be presented as these calibrated research

probabilities.

Betting returns

Gross BSP P/L is the default. Industry SP settlement remains separate. Stakes come from each frozen book, non-runners void, dead heats use the recorded divisor, and **RR/refused to race is a loser**, not a non-runner. Missing/unresolved results are explicitly unresolved, not silently zero-return days. Runner matching and freeze/settlement joins must be checked before quoting totals.

Higher P/L over one sample can coexist with worse probability accuracy. Large odds winners, changed bet counts, overlapping books and repeated trials can dominate returns. Compare the same races, inspect common nominated races, report months and uncertainty, and show largest-winner sensitivity without selecting rules afterward.

The 527-race repeated historical comparison is not an untouched holdout. Registered prospective rules, pre-off forecast capture and separate later outcomes are the stronger next evidence. A blocked or unsealed date cannot be called prospective simply because a prediction file exists.

11. Today's concrete example

The 11 October final followed CSV contains 5 horses /13pt:4Flat and 1 Jumps. The delivered shadow contains 7 horses /7separate points:2 Flat and 5 Jumps. These counts verify the description, not a new recommendation or changed card.

At Goodwood 13:47, Carthusian qualifies through PR 1+Production 1 and Klycot through Benter No-Current. They are different followed lanes and the final 2-per-race guard allows both.

At Chepstow 14:02, Ugo Bingo and Little Pi are Jumps-shadow model ranks 1 and 2 with qualifying source ranks; both remain. The same applies to Norn Iron/Gower Prince at 16:22. Flat-shadow races with two qualifying model-top-two horses are excluded instead. This is why several horses in one race are not necessarily a duplicate or a malfunction.

12. What this means operationally

1. Preserve the current followed system and delivered shadow exactly unless a dated rule/model change is explicitly promoted.

2. Assess full-field forecasting and paper returns separately.

3. Keep new past-performance/conditions ideas in named, frozen research versions.

4. Count verified prospective captures, not planned days or successful software tests, as live evidence.

5. Do not claim that “we tested it” means “we now use it in daily selections”.

The most important open limitations are probability calibration, modest clean monthly training samples, incomplete historical prediction-time coverage, experimental achieved-rating assumptions and research operational capture failures. More features do not automatically solve these.

Appendix A. Current delivered Step-1 feature dictionary

The generated table below lists the exact October frozen model feature names. All 48 are in Flat; Jumps excludes dynamic_speed. Definitions describe the inspected feature-building code, not a claim of statistical importance.

Feature	Meaning	Flat	Jumps
Age	Declared horse age.	Yes	Yes
Pace	Source pace/run-style number; not measured RaceIQ sections.	Yes	Yes

Feature	Meaning	Flat	Jumps
Class	Declared race class.	Yes	Yes
Runners	Source field size.	Yes	Yes
Prev Races	Source count of earlier races.	Yes	Yes
Days Since LTO	Days since previous run.	Yes	Yes
WGT (Lbs)	Declared weight in pounds; not a verified net-claim weight.	Yes	Yes
OR	Official rating/handicap mark.	Yes	Yes
OR diff since LTO	Source change in OR since latest run.	Yes	Yes
Cla diff since LTO	Source class change since latest run.	Yes	Yes
Dist (F)	Declared distance in furlongs.	Yes	Yes
industry_prob	Legacy name: 1 / validated RTR morning odds in daily shadow.	Yes	Yes
field_or_mean	Mean OR in this field.	Yes	Yes
field_or_max	Maximum OR in this field.	Yes	Yes
or_vs_mean	Horse OR minus field mean.	Yes	Yes
or_vs_max	Horse OR minus field maximum.	Yes	Yes
or_rank_in_race	OR rank, highest first; average ties.	Yes	Yes
field_weight_mean	Mean declared weight in this field.	Yes	Yes
weight_vs_mean	Declared weight minus field mean.	Yes	Yes
weight_rank_in_race	Weight rank, lowest first; average ties.	Yes	Yes
field_market_prob_sum	Sum of raw implied market probabilities.	Yes	Yes
market_rank_in_race	Raw implied probability rank, highest first.	Yes	Yes
market_prob_gap_to_fav	Favourite implied probability minus horse implied probability.	Yes	Yes
market_top2_share	Raw sum of the two highest implied probabilities; not normalized share.	Yes	Yes
market_top3_share	Raw sum of the three highest implied probabilities; not normalized share.	Yes	Yes
field_pace_mean	Mean source Pace.	Yes	Yes
pace_vs_mean	Source Pace minus field mean.	Yes	Yes
pace_rank_in_race	Source Pace rank, highest first.	Yes	Yes
front_runner_count	Number of source Pace values at least 7.	Yes	Yes
field_age_mean	Mean field age.	Yes	Yes
age_vs_mean	Age minus field mean.	Yes	Yes
field_prev_races_mean	Mean prior-race count of field.	Yes	Yes
prev_races_vs_mean	Prior-race count minus field mean.	Yes	Yes
field_days_mean	Mean days since latest run in field.	Yes	Yes
days_vs_mean	Days since latest run minus field mean.	Yes	Yes
Track	Course identity category.	Yes	Yes
Going	Prediction-time going category.	Yes	Yes

Feature	Meaning	Flat	Jumps
Racetype	Race-type category.	Yes	Yes
Jockey	Jockey identity category, not explicit strike rate.	Yes	Yes
Trainer	Trainer identity category, not explicit strike rate.	Yes	Yes
Headgear	Declared equipment category.	Yes	Yes
Course Winner	Source course-winner summary flag.	Yes	Yes
Distance Winner	Source distance-winner summary flag.	Yes	Yes
Up in Trip	Source trip-change flag.	Yes	Yes
Country	Country category; eligible shadow scope is GB.	Yes	Yes
pr_score_log	log1p of nonnegative prediction-time Racedata PR Score.	Yes	Yes
Racedata__PR Rank	Prediction-time PR rank.	Yes	Yes
dynamic_speed	Raw prediction-time Racedata Dynamic speed, Flat only.	Yes	No

Appendix B. Legacy Jumps suitability inputs

LTO Pos; Wins L 5; Runs last 18 mo; Dist Wins; Crs Wins; Going Wins; Days Since LTO; Prev Races; Course Winner; Distance Winner; Up in Trip; Going; Track; Racetype; Class; Dist(F); Age; ORdiffsinceLTO; CladiffsinceLTO; going_specialist; distance_specialist; course_specialist; activity_ratio; recency_score; form_score.

Specialist/activity ratios divide the relevant summary count by prior races (with a safe positive denominator). recency_score=exp(−days/30). form_score=last-five wins plus a latest finishing-position bonus of 1 for first,0.5 for second or 0.25 for third. These summaries are not baseline-relative condition residuals.

Appendix C. Sources, versions and verification

The accompanying SOURCE_AUDIT.json records read-source hashes, model identities and document checks. Main references:

- reports/source_completeness/2026-10-11/prerace_release.json

- reports/daily_handicap_shadow/2026-10-11/combined_followed_with_blend_2026-10-11.csv

- reports/daily_handicap_shadow/2026-10-11/rolling_rich_step1_shadow_provenance_2026-10-11.json

- reports/rich_step1_prospective_shadow/seals/2026-10-11/model_manifest.json and speed_model_manifest.json

- reports/rich_step1_prospective_shadow/PRICE3_AMENDMENT_2026-10-06.md

- tools/build_daily_handicap_shadow_lists.py; tools/followed_rtr_price_policy.py

- tools/build_daily_bred_signal.py; tools/build_daily_top_combo.py; tools/build_daily_weighted.py

- research/benter_value_engine/build_daily_no_current_overlay.py

- reports/research_handicap_universe_workflow.py; research_handicap_universe_noprice_workflow.py; research_jumps_baseline_workflow.py; research_jumps_blend20_paper_workflow.py

- reports/step1_tournament_v1/REGISTRATION.json, STATUS.json and operations.jsonl

- reports/step1_rating_prospective_v1/REGISTRATION.json

- reports/draw_pace_context_trial_v1/CURRENT_STATUS.json

- reports/horse_residual_suitability/2026-10-11/REPORT.md; larger_2026_v1/REPORT.md

This is a dated operational explanation, not a retrospective certification of every legacy training row or every past rule. Where old comments/manifests conflict with current dated selection provenance, the distinction is recorded rather than hidden.